

Module 3 – Docker

1. Docker Introduction

What is Docker?

Docker is a platform used to create, run, and manage applications inside containers.

Why Docker?

- Consistent environment
- Lightweight compared to virtual machines
- Easy deployment

Key Idea

“Build once, run anywhere”

2. Docker Images

What is a Docker Image?

A Docker image is a blueprint used to create containers.

Features

- Contains application code
- Includes dependencies
- Read-only template

Example:

```
docker build -t myapp .
```

3. Docker Containers

What is a Container?

A container is a running instance of a Docker image.

Benefits

- Fast startup
- Isolated environment
- Efficient resource usage

Example:

```
docker run -d -p 3000:3000 myapp
```

4. Docker Compose

What is Docker Compose?

Docker Compose is a tool used to run multiple containers together.

Why Use It

- Manage multi-service apps
- Simplify configuration
- Easy startup with one command

Example:

```
version: "3"
services:
  app:
    build: .
    ports:
      - "3000:3000"
```

5. Best Practices

Important Guidelines

- Use small base images
 - Avoid unnecessary files
 - Use .dockerignore
 - Keep images secure
 - Use multi-stage builds
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6. Key Concepts to Remember

- Docker uses containers for applications
 - Images act as blueprints
 - Containers run the app
 - Compose manages multiple containers
 - Best practices improve performance
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7. Summary

Docker simplifies application deployment by packaging code and dependencies into containers. It ensures consistency across environments and makes applications easier to scale and manage.